

Lab Experiment 9: Create, Delete, and Modify Local Groups and Group Memberships

Objective

To understand how to **create**, **delete**, and **modify** local groups in Linux, and how to **add or remove users** from those groups.

Background Concept

In Linux, a **group** is a collection of users that share common permissions. Groups help manage **file access** and **system privileges** easily — instead of assigning permissions to each user individually.

Every user belongs to:

- **A primary group** (created automatically when the user is added)
- **One or more secondary groups** (added manually as needed)

Basic Terms

Term	Meaning
Primary Group	The main group assigned to a user at creation
Secondary Group	Additional groups the user can be a member of
/etc/group	File that stores information about groups
Group Membership	The list of users that belong to a group

Step-by-Step Procedure

Step 1: Open Terminal and Switch to Root

To perform administrative tasks:

```
sudo su
```

Step 2: View Existing Groups

To see all groups on the system:

```
cat /etc/group
```

You'll see lines like:

```
students:x:1001:  
faculty:x:1002:trainer
```

Here, **faculty** is the group name, and **trainer** is a member of it.

Step 3: Create a New Group

To create a new group:

```
sudo groupadd projectteam
```

To verify creation:

```
getent group projectteam
```

Step 4: Add a User to a Group

Add an existing user (e.g., **student1**) to a group (e.g., **projectteam**):

```
sudo usermod -aG projectteam student1
```

Note: The **-aG** option means “append user to supplementary group(s)” — very important to avoid removing them from other groups.

To check the user's group memberships:

```
groups student1
```

Step 5: Create a User with a Specific Group

You can directly assign a primary group when creating a user:

```
sudo useradd -g projectteam student2  
sudo passwd student2
```

Step 6: Change a User's Primary Group

To change the main group of a user:

```
sudo usermod -g faculty student1
```

Check:

```
id student1
```

It will show updated primary group info.

Step 7: Remove a User from a Group

To remove a user from a secondary group:

```
sudo gpasswd -d student1 projectteam
```

Check using id student1

Step 8: Modify Group Information

To change a group name:

```
sudo groupmod -n developers projectteam
```

Check using group student2

To change group ID (GID): first check using id student2

```
sudo groupmod -g 1050 developers
```

Verify using id student2

Step 9: Delete a Group

To remove a group completely:

```
sudo groupdel developers
```

Verify deletion:

```
getent group developers
```

(No output means the group is deleted.)

Practice Tasks (Recommended for Students)

1. Create three groups: `team1`, `team2`, and `adminteam`.
2. Create a user named `user1` and assign it to `team1`.
3. Add `user1` to `team2` as a secondary group.
4. Display all groups to which `user1` belongs.
5. Rename `team2` to `qa_team`.
6. Remove `user1` from `qa_team`.
7. Create another user `user2` and assign its primary group as `adminteam`.
8. View `/etc/group` file to confirm all changes.
9. Delete the `qa_team` group.
10. Verify the deleted group no longer appears in `/etc/group`.

Summary

Command	Description
<code>groupadd</code>	Create a new group
<code>groupdel</code>	Delete a group
<code>groupmod</code>	Modify a group (name or ID)
<code>usermod -aG</code>	Add user to supplementary group
<code>gpasswd -d</code>	Remove user from group
<code>id / groups</code>	Check user's group membership

Learning Outcome

By completing this experiment, students will be able to:

- Understand the concept of groups and group memberships.
- Create, delete, and modify groups.
- Manage user-group relationships effectively.
- View and verify group configurations using basic Linux commands.